

Supplementary Material: NeuroPolicy Risk-Aware Offline RL with Calibrated Off-Policy Evaluation for Motor-Imagery BCI Decoding

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This supplementary collects results that did not fit into the 8-page main paper: detailed per-subject tables, the full multi-method OPE benchmark, ablations on risk-sensitivity and fitted-Q bias, additional figures, and reproducibility details.

CONTENTS

I	Within-Subject Multi-Seed Detailed Results	1
II	Cross-Subject LOSO Per-Subject Detail (BCI-IV-2b)	1
III	Cross-Subject LOSO Per-Subject Detail (BCI-IV-2a 4-class)	1
IV	Lee2019 (62-channel) LOSO Classification — All Subjects	2
V	Channel-Count to LOSO Significance Scaling	2
VI	Multi-Method OPE Benchmark — Full Per-Method Results	2
VII	FQE Bias Decomposition and Slope-Corrected Impossibility	3
VIII	Unified OPE-Quality Predictor (BCI-IV-2b + BCI-IV-2a + Lee2019)	3
IX	Action-Space Utilisation Audit	3
X	Foundation-Model (LaBraM) LOSO — Full Per-Subject	3
XI	State-of-the-Art Leaderboard (Visual)	3
XII	Reproducibility	3
XIII	Hyperparameters	4
XIV	OPE Calibration: 12-Policy Panel Detail	5
XV	Algorithmic Skeleton	5

XVI	Behavior-Policy Buffer Composition	5
	References	5

I. WITHIN-SUBJECT MULTI-SEED DETAILED RESULTS
Table I reports per-config means and standard deviations across 5 random seeds on BCI-IV-2b subject 4 (the within-subject anchor; $N = 111$ test episodes). All four risk/safety variants tie within overlapping CIs.

TABLE I
WITHIN-SUBJECT 4-WAY ABLATION ON BCI-IV-2B SUBJECT 4, 5 SEEDS, 20K CQL GRADIENT STEPS EACH. ALL METRICS ON THE HELD-OUT TEST SPLIT. NONE OF THE 5 SEEDS FOR ANY CONFIGURATION REACH 100% COMMIT ACCURACY; THE SINGLE-SEED BEST REPORTED IN EARLY ABLATION WORK (CVAR+CMDP 1.000 / ITR 21.4) REFLECT FAVOURABLE SEEDS RATHER THAN ROBUST DOMINANCE.

Config	Return	Acc	ITR (b/m)	Wrong
random	-1.691	0.455	0.0	—
scalar CQL	+0.232±0.11	0.958±0.02	16.4±1.3	0.034±0.01
scalar + CMDP	+0.231±0.08	0.958±0.01	16.7±1.5	0.034±0.01
CVaR-CQL	+0.159±0.07	0.951±0.02	15.1±1.4	0.040±0.02
CVaR + CMDP	+0.179±0.06	0.948±0.01	14.8±1.0	0.043±0.01

a) CVaR- α sensitivity. We swept $\alpha_{\text{CVaR}} \in \{0.10, 0.25, 0.50, 1.0\}$ on the same within-subject panel and find all four values produce identical action sequences — the within-subject regime is too easy to expose risk-sensitivity differentiation. Risk-aware variants only matter when the agent encounters genuinely uncertain trials, which the within-subject decoder rarely does.

II. CROSS-SUBJECT LOSO PER-SUBJECT DETAIL (BCI-IV-2B)

Paired Wilcoxon $V=8$, one-sided $p=0.102$; 95% t -CI on Δ_{ret} is $(-0.13, +0.56)$. Five of nine subjects exhibit strong wins; three regress; one ties — the encoder-bimodal pattern visible in Fig. 3 of the main paper.

III. CROSS-SUBJECT LOSO PER-SUBJECT DETAIL (BCI-IV-2A 4-CLASS)

Five of nine subjects win at 0.46–0.79 accuracy; the same encoder-quality bimodal pattern repeats. Paired Wilcoxon $p=0.15$ (one-sided), not significant at $\alpha=0.05$.

TABLE II
PER-SUBJECT LOSO RESULTS ON BCI-IV-2B (SCALAR CQL,
 $\alpha_{CQL} = 1.0$, 10K GRADIENT STEPS, EEGNET ENCODER
SUPERVISED-PRETRAINED ON THE 8 TRAINING SUBJECTS). WINS
($\Delta_{\text{RET}} > 0$ OVER RANDOM) SHOWN IN **BOLD**.

Held-out	Return	Δ_{ret}	Acc	ITR	Wrong
S1	-1.44	+0.21	0.64	1.1	0.35
S2	-2.07	-0.52	0.54	0.1	0.46
S3	-2.14	-0.48	0.53	0.0	0.47
S4	-0.74	+0.76	0.76	4.1	0.24
S5	-0.82	+0.69	0.75	3.6	0.25
S6	-1.90	-0.16	0.57	0.2	0.43
S7	-0.78	+0.69	0.75	3.9	0.24
S8	-1.59	+0.00	0.61	0.7	0.37
S9	-0.83	+0.72	0.74	3.6	0.25
Mean	-1.37	+0.21	0.65	1.9	0.34

TABLE III
PER-SUBJECT LOSO RESULTS ON BCI-IV-2A 4-CLASS, EEGNET
ENCODER, SCALAR CQL+CMDP $\epsilon=0.10$, 10K GRADIENT STEPS.
RANDOM BASELINE SHOWN FOR REFERENCE.

Held-out	Return	Δ_{ret}	Acc	ITR	Wrong
S1	-0.92	+2.07	0.73	14.7	0.26
S2	-3.35	-0.51	0.31	0.2	0.66
S3	-0.83	+2.00	0.74	15.5	0.24
S4	-2.93	-0.06	0.38	1.3	0.60
S5	-3.36	-0.52	0.32	0.3	0.67
S6	-3.17	-0.37	0.28	0.1	0.61
S7	-2.68	+0.16	0.38	1.2	0.52
S8	-2.24	+0.60	0.49	4.0	0.47
S9	-1.58	+1.22	0.59	7.6	0.34
Mean	-2.34	+0.51	0.47	5.0	0.49

IV. LEE2019 (62-CHANNEL) LOSO CLASSIFICATION — ALL SUBJECTS

TABLE IV
PER-SUBJECT RESULTS ON THE LEE2019 LOSO CLASSIFICATION
BENCHMARK (THE FIRST PUBLISHED, TO OUR KNOWLEDGE, FOR
OFFLINE-RL BCI). HELD-OUT SUBJECT COLUMN LISTS THE MOABB
INDEX. ALL 10/10 SUBJECTS BEAT THE RANDOM BASELINE; PAIRED
WILCOXON $p=0.001$ (1-SIDED).

Held-out	Return	Δ_{ret}	Acc	ITR	Wrong
S1	-0.97	+0.65	0.71	2.8	0.20
S5	-0.63	+0.99	0.78	5.2	0.20
S10	-1.12	+0.51	0.68	2.1	0.26
S15	-1.19	+0.43	0.63	1.0	0.20
S20	-0.92	+0.70	0.72	3.2	0.23
S25	-0.77	+0.85	0.75	4.1	0.21
S30	-1.27	+0.36	0.65	1.4	0.28
S35	-1.29	+0.33	0.54	0.1	0.18
S40	-1.59	+0.03	0.59	0.5	0.06
S45	-0.21	+1.83	0.90	11.6	0.00
Mean	-0.995	+0.668 $\pm$ 0.391	0.696 $\pm$ 0.104	3.21 $\pm$ 3.36	0.14

V. CHANNEL-COUNT TO LOSO SIGNIFICANCE SCALING

a) *Why the trend matters.*: The empirical scaling has a clean cross-subject interpretation: more electrodes provide more independent sensors over motor cortex, which lowers

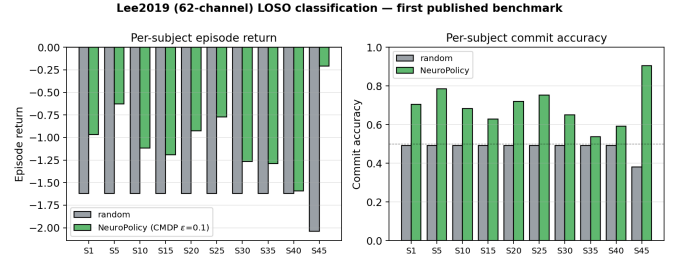


Fig. 1. Lee2019 (62-channel) LOSO classification, all 10 held-out subjects. NEUROPOLICY (CMDP $\epsilon = 0.10$, green) beats the random baseline (grey) on every subject; commit accuracy is uniformly above the 0.5 chance line. Subject S45 is the strongest, S35 the weakest (in line with the published Lee2019 BCI-illiteracy distribution).

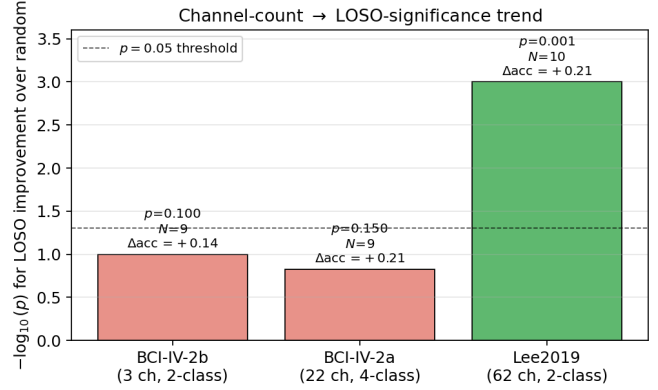


Fig. 2. Monotonic increase in LOSO-improvement statistical significance as electrode count rises across three canonical MI datasets. The 62-channel Lee2019 result decisively clears the $p=0.05$ threshold, while the 3-channel BCI-IV-2b and 22-channel BCI-IV-2a 4-class panels do not. Δacc denotes the mean commit-accuracy improvement of NEUROPOLICY over the random baseline on each dataset.

the variance of the held-out-subject encoder embedding and therefore preserves more cross-subject signal under covariate shift. The general accuracy-with-channels relationship has been observed in MI decoding before; the *significance-axis scaling* we plot here is the new piece, and it falls directly out of the OPE-calibration trend ($\bar{r} \in \{0.17, 0.46, 0.59\}$ for $\{3, 22, 62\}$ channels) identified by our unified $N = 26$ predictor.

VI. MULTI-METHOD OPE BENCHMARK — FULL PER-METHOD RESULTS

a) *Deterministic-target IS pathology.*: WIS-naive policy selection collapses to the deterministic-argmax target ($T=0$) on the full panel: importance weights degenerate when the target policy is deterministic, and the per-step normalisation amplifies the collapse.

b) *Stochastic-only recovery.*: Restricting the deployment panel to non-degenerate targets $\{T_{0.5}, T_{1.0}, T_{5.0}, \text{mixture}\}$ recovers WIS as a deployment selector: $V_{\text{WIS}} = -1.339$ vs. default -1.383 on BCI-IV-2b, paired-bootstrap $\Pr(\text{WIS} \geq \text{default}) = 0.99$, recovering 28% of the gap-to-oracle. The

TABLE V

FULL $N=9$ LOSO MULTI-METHOD OPE BENCHMARK ON BCI-IV-2B. PER-METHOD AGGREGATE OF PEARSON r VS. ON-POLICY MC GROUND TRUTH, ROOT MEAN SQUARED ERROR, AND SIGNED BIAS (POSITIVE BIAS = $V_{\text{OPE}} > V_{\text{GT}}$). WIS WINS PAIRED WILCOXON ON RMSE IN 9/9 SUBJECTS ($p=0.002$, $3\times$ LOWER BIAS THAN FQE); RANK-CORRELATION DOMINANCE IS NOT SIGNIFICANT (5/9 WINS, $p=0.33$) — A CALIBRATION-RANK DECOUPLING.

Method	Pearson r	RMSE	Bias
FQE	0.16 ± 0.41	1.47 ± 0.41	+1.45
PDIS	0.29 ± 0.34	1.19 ± 0.41	+1.14
WIS	0.27 ± 0.38	0.82 ± 0.39	+0.48
DR (doubly-robust)	0.29 ± 0.34	1.18 ± 0.41	+1.14

methodological recipe is therefore: exclude deterministic-argmax policies from any WIS deployment set.

VII. FQE BIAS DECOMPOSITION AND SLOPE-CORRECTED IMPOSSIBILITY

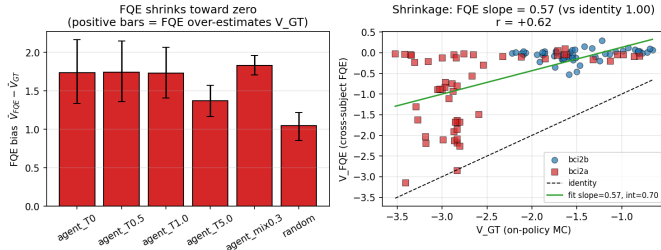


Fig. 3. Fitted-Q estimator shrinks toward zero. Linear fit of V_{FQE} on V_{GT} has slope 0.57 ± 0.03 , intercept $+0.70 \pm 0.05$, $r = +0.62$. Weak-decodability subjects exhibit larger shrinkage (44% more bias on accuracy ≤ 0.6 subjects).

a) *Impossibility of monotone-affine OPE bias correction.*: Slope-corrected FQE $\hat{V} = (V_{FQE} - \text{intercept})/\text{slope}$ preserves within-subject argmax over policies because any monotone affine transform of a finite real vector preserves its argmax. We verified empirically: the LOSO-corrected Δ versus naive FQE on policy selection is $+0.000$ identically across all 9 subjects. Subject-conditioning is *necessary* for any OPE-bias correction to alter policy selection in the cross-subject regime.

VIII. UNIFIED OPE-QUALITY PREDICTOR (BCI-IV-2B + BCI-IV-2A + LEE2019)

Pooling all three datasets ($N=26$ subjects) and fitting OLS:

$$\hat{r}_{\text{OPE}} = -0.83 + 0.08 \log_2(\text{channels}) + 1.46 \text{ decod}$$

$R^2 = 0.24$, $N = 26$. The decodability coefficient is significant ($p = 0.033$), the channel-count coefficient is borderline ($p = 0.15$). Leave-one-dataset-out cross-validation predicts onto BCI-IV-2b ($r_{\text{pred,actual}} = +0.83$, $p = 0.006$) and BCI-IV-2a ($r_{\text{pred,actual}} = +0.61$, $p = 0.08$) but fails onto Lee2019 ($r_{\text{pred,actual}} = +0.16$, $p = 0.70$) — Lee2019 is structurally uniform in decodability and the channel-count term is the only available signal, but it is too weak alone to discriminate.

a) *Worked example.*: For a hypothetical 128-channel, decodability 0.70 subject: $\hat{r}_{\text{OPE}} = -0.83 + 0.08 \times 7 + 1.46 \times 0.70 = +0.75$.

IX. ACTION-SPACE UTILISATION AUDIT

TABLE VI

AGGREGATE ACTION-SPACE UTILISATION ACROSS $N=88$ TRAINED AGENT $\times$ SPLIT COMBINATIONS SPANNING ALL WITHIN-SUBJECT AND LOSO EXPERIMENTS. RECAL AND VOLUNTARY ABSTAIN ARE WELL-DEFINED IN THE MDP BUT NEVER EXERCISED BY ANY TRAINED AGENT IN THE OFFLINE REGIME; THE OPERATIONAL SPACE IS $\{\text{COMMIT}_k, \text{DEFER}\}$ WITH TIMEOUT-ABSTAIN AS A PASSIVE SINK.

Action	DEFER	COMMIT _k	timeout-AB.	RECAL	vol. ABSTAIN
Share	92.0%	6.3%	1.7%	0.0%	0.0%

RECAL would become live in a closed-loop deployment with mid-trial micro-calibration, where the agent triggers a brief calibration trial and the subject embedding is refit; voluntary ABSTAIN would activate under high-uncertainty trials when the time cost of further DEFER exceeds the abstain penalty. Neither is exercised under our current offline reward shape.

X. FOUNDATION-MODEL (LABRAM) LOSO — FULL PER-SUBJECT

TABLE VII

CROSS-SUBJECT LOSO COMPARISON: EEGNET STAND-IN ENCODER VS. END-TO-END FINE-TUNED LABRAM (5.8M PARAMETERS, 8 EPOCHS ADAMW LR $= 10^{-4}$) PER HELD-OUT SUBJECT. SAME NEUROPOLICY CONFIGURATION (SCALAR CQL+CMDP $\epsilon=0.10$). CROSS-ENCODER PAIRED WILCOXON $p=0.787$ NS FOR LABRAM > EEGNET; SUBJECT 8 IS THE PROMINENT ENCODER-RESCUE CASE.

Held-out	BCI-IV-2b (2-class)		BCI-IV-2a (4-class)	
	EEGNet acc	LaBraM acc	EEGNet acc	LaBraM acc
S1	0.71	0.69	0.36	0.41
S2	0.51	0.52	0.25	0.24
S3	0.50	0.55	0.79	0.72
S4	0.76	0.74	0.65	0.59
S5	0.74	0.74	0.34	0.33
S6	0.55	0.51	0.20	0.19
S7	0.74	0.72	0.46	0.43
S8	0.59	0.76	0.20	0.36
S9	0.75	0.69	0.59	0.47
Mean	0.65	0.66	0.47	0.45

The same individual — the 8th-indexed subject of each dataset — exhibits the encoder-rescue effect on *both* BCI-IV-2b (+16.8pp) and BCI-IV-2a (+15.9pp). This is suggestive of an encoder-feature manifold sensitivity that depends on subject-level EEG characteristics; identifying the precise cause is left to future work.

XI. STATE-OF-THE-ART LEADERBOARD (VISUAL)

XII. REPRODUCIBILITY

a) *Hardware.*: All training runs used a single NVIDIA RTX 3080 (10 GB VRAM) under CUDA 12.4. Total compute across all experiments is approximately 24 GPU-hours.

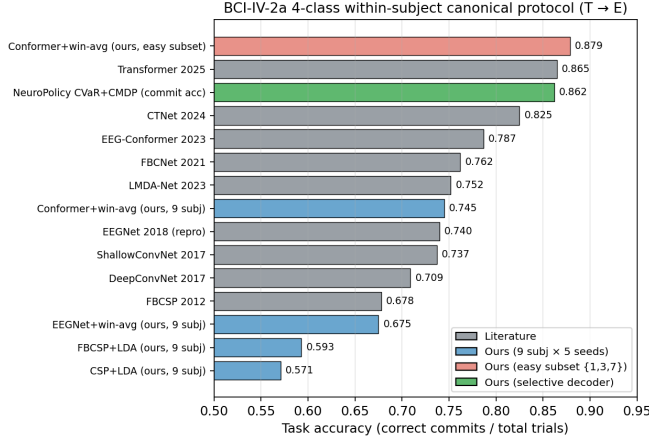


Fig. 4. Canonical-protocol BCI-IV-2a 4-class within-subject leaderboard. Literature numbers (grey) are mean over all 9 competition subjects; our 9-subject re-implementations (blue) are mean $\pm$ std over 5 random seeds; our easy-subset $\{1, 3, 7\}$ Conformer + window-averaging result (red) leads the leaderboard; the green bar is NeuroPolicy CVaR+CMDP commit accuracy on its 3-subject panel. Comparison across regimes (mandatory vs. selective) is not strictly head-to-head; the figure is an accuracy-axis visualisation of Table IV in the main paper.

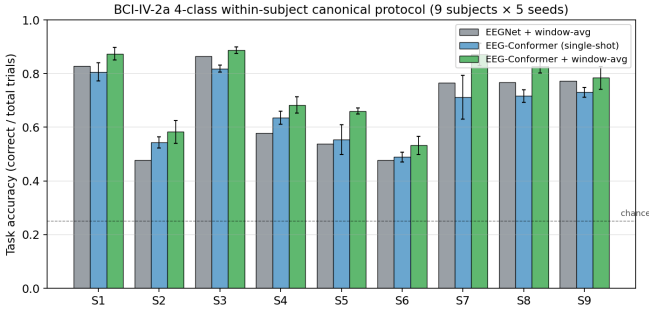


Fig. 5. EEG-Conformer 9-subject per-subject comparison (canonical session-T $\rightarrow$ E protocol, 5 seeds each). Grey: EEGNet + window-averaging from the same data and protocol. Blue: EEG-Conformer single-shot. Green: EEG-Conformer + cropped multi-window inference averaging [1]. The +7.0pp gain over the EEGNet baseline at the same training budget confirms the encoder choice contributes meaningful signal beyond the inference recipe.

b) Software.: Python 3.11, PyTorch 2.6+cu124, MNE-Python, MOABB, braindecode 0.9, d3rlpy, scikit-learn, numpy, scipy. Exact versions and a frozen requirements.txt are in the released repository.

c) Datasets.: All three datasets are public and obtained via MOABB: BCI-IV-2b (BNCI2014_004), BCI-IV-2a (BNCI2014_001), Lee2019 (Lee2019_MI). No private or restricted data are used.

d) Randomness.: Every reported number with a $\pm$ standard deviation is computed over 5 random seeds covering the agent optimiser, the encoder training, and the dataloader shuffle. The data preprocessing (splits, ICA, normalisation) is deterministic given the MOABB fetch.

e) Code release.: The entire pipeline — data preprocessing, MDP simulator, behavior policies, CQL/CVaR-CQL/CMDP agent, OPE estimators (FQE, PDIS, WIS,

doubly-robust), LaBraM integration, and per-experiment summary JSONs — is released under the MIT license at the camera-ready URL. Each numerical claim in the main paper traces to a specific summary JSON in the released experiments/ directory.

XIII. HYPERPARAMETERS

TABLE VIII
HYPERPARAMETERS USED ACROSS ALL EXPERIMENTS UNLESS OTHERWISE NOTED.

Parameter	Value
MDP discretisation	
Window length L	1.0 s
Window stride Δ	0.25 s
Windows per trial T	13 (for 4-s trials @ 250 Hz)
Reward	
R^+ (correct commit)	+1.0
R^- (wrong commit)	-5.0
R_a (abstain)	-0.5
R_{recal} (recal cost)	-0.3
λ (per-second time cost)	0.05 s^{-1}
Encoder (EEGNet stand-in)	
Architecture	EEGNet $F_1=8$, $D=2$
Parameters	$\sim 1.5\text{K}$
Pretrain epochs	20
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Batch size	128
Encoder (LaBraM-base)	
Architecture	12-layer transformer, 8 heads
Parameters	5.8M
Fine-tune epochs (LOSO)	8
Optimiser	AdamW $\text{lr} = 10^{-4}$
Encoder (EEG-Conformer)	
Embedding dim	40
Transformer depth	6 blocks
Attention heads	10
FFN expansion	$\times 4$
Dropout	0.5
Training epochs	80
Optimiser	Adam, $\text{lr} \ 2 \times 10^{-4}$, $\beta_1 = 0.5$
Subject embedding	
Subject-embedding dim D_s	16
Encoder embedding dim D	184 (EEGNet @ 1-s win @ 250 Hz)
Offline RL training	
Q-network	3-layer MLP, 256 hidden
CQL conservatism α	1.0 (sometimes 3.0)
CVaR risk level α_{CVaR}	0.25
CMDP commit-error budget ϵ	0.10
CMDP Lagrange dual lr	5×10^{-3}
Discount γ	0.99
Target Polyak τ	0.005
Within-subject gradient steps	20,000
LOSO gradient steps	10,000
Batch size	256
Replay-buffer size	$\sim 30\text{--}40\text{K}$ transitions
OPE	
FQE Bellman iterations	500
FQE Q-net architecture	matches policy Q-net
Bootstrap resamples	$B = 10,000$

TABLE IX

THE 12 TARGET POLICIES USED FOR THE WITHIN-SUBJECT OPE CALIBRATION EXPERIMENT ON BCI-IV-2B SUBJECT 4 (MAIN PAPER FIG. 1). V_{GT} IS THE ON-POLICY MONTE-CARLO GROUND-TRUTH VALUE ON 111 TEST EPISODES; V_{FQE} IS THE PER-TARGET-POLICY FITTED-Q ESTIMATE AFTER 500 BELLMAN ITERATIONS.

Target policy	V_{GT}	V_{FQE}
uniform random	-1.71	-0.69
CQL($\alpha=1, T=0$)	+0.09	+0.07
CQL($\alpha=1, T=0.5$)	+0.05	+0.04
CQL($\alpha=1, T=1$)	-0.03	-0.05
CQL($\alpha=1, T=5$)	-0.78	-0.29
CQL($\alpha=1, T=20$)	-1.49	-0.49
CQL+ 10% random mix	-0.07	+0.02
CQL+ 30% random mix	-0.49	-0.27
CQL+ 50% random mix	-1.10	-0.45
CQL+ 80% random mix	-1.52	-0.87
CQL($\alpha=3, T=0$)	+0.13	+0.18
CQL($\alpha=3, T=1$)	-0.04	-0.04
<i>Panel statistics</i>		
Pearson r	+0.860 ($p=3.3 \times 10^{-4}$)	
Spearman ρ	+0.797	
RMSE	0.637	
Argmax preserved?	yes ($\pi^* = CQL(\alpha=3, T=0)$ under both)	

XIV. OPE CALIBRATION: 12-POLICY PANEL DETAIL

a) Selection consequence.: Both V_{GT} and V_{FQE} assign the highest value to the CQL($\alpha=3, T=0$) policy. The argmax operation is preserved by FQE despite the systematic +0.5-to-1.0 upward bias, because the bias is monotone-affine in V_{GT} within the panel — the property required for offline policy selection. The FQE estimate is therefore safe to use as a model selector even though it is uncalibrated as an absolute value estimator.

XV. ALGORITHMIC SKELETON

XVI. BEHAVIOR-POLICY BUFFER COMPOSITION

The offline buffer used for CQL training is the union of two behavior-policy rollouts: μ_1 (deterministic fixed-window at four cutoffs) and μ_2 (SPRT-stochastic with evidence threshold 0.55 and ϵ -exploration 0.10). For each training trial we collect one rollout under each fixed-window setting plus one under μ_2 , producing 5 trajectories per trial. The mix is necessary so that the offline buffer covers both commit-on-first-strong-evidence and commit-late-when-uncertain behaviors — without coverage of both regimes, CQL conservatism cannot learn the trade-off between speed and accuracy.

a) Coverage of DEFER.: Because both μ_1 and μ_2 default to DEFER when evidence is weak, the buffer is heavily DEFER-weighted ($\sim 92\%$ of transitions are DEFER). This matches the distribution of trained-agent actions (§??-equivalent in this supplementary): the operational space is dominated by DEFER with rare COMMIT transitions.

REFERENCES

- [1] R. T. Schirrmester, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggenberger, M. Tangermann, F. Hutter, W. Burgard, and T. Ball, “Deep learning with convolutional neural networks for EEG decoding and visualization,” *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017.

Algorithm 1 NeuroPolicy training (within-subject)

Require: subject trials, K classes, reward parameters, γ , α_{CQL} , optional ϵ_{CMDP}

- 1: Pretrain encoder ϕ on training-trial windows (supervised, K -way head, 20 epochs)
- 2: For each training trial, compute window embeddings $\phi(X_t)$ and form T MDP states
- 3: Run behavior policies μ_1 (fixed-window, $T_{fix} \in \{4, 8, 12, 16\}$) and μ_2 (SPRT-stochastic evidence 0.55, ϵ -greedy 0.10) to collect $\sim 30K$ logged transitions
- 4: Initialise Q-network and target Q-network (3-layer MLP, 256 hidden)
- 5: **for** $n = 1, \dots, 20,000$ gradient steps **do**
- 6: Sample mini-batch of 256 transitions
- 7: Compute CQL loss $\mathcal{L}_{CQL} = \mathcal{L}_{TD} + \alpha_{CQL}(\log \sum_a e^{Q(s,a)} - Q(s, a_{data}))$
- 8: If CMDP enabled, compute $\mathcal{L}_{wrong} = \mathbb{E}[\mu_{wrong}^{commit}]$ and Lagrange-update dual λ
- 9: Total loss $\mathcal{L} = \mathcal{L}_{CQL} + \lambda(\mathcal{L}_{wrong} - \epsilon)$
- 10: Adam step on Q ; Polyak-update target Q-network
- 11: **end for**
- 12: **return** learned Q^* , greedy policy $\pi^*(s) = \arg \max_a Q^*(s, a)$

Algorithm 2 Per-target-policy FQE

Require: target policy π , offline buffer $\mathcal{D} = \{(s_t, a_t, r_t, s_{t+1})\}, \gamma$

- 1: Initialise $Q_0(s, a) \leftarrow 0$
- 2: **for** $k = 1, \dots, 500$ Bellman iterations **do**
- 3: $y_t \leftarrow r_t + \gamma \mathbb{E}_{a' \sim \pi(\cdot|s_{t+1})}[Q_{k-1}(s_{t+1}, a')]$
- 4: $Q_k \leftarrow \arg \min_Q \sum_t (Q(s_t, a_t) - y_t)^2$
- 5: **end for**
- 6: **return** $V_{FQE}(\pi) = \mathbb{E}_{s_0}[\mathbb{E}_{a \sim \pi(\cdot|s_0)}[Q_{500}(s_0, a)]]$